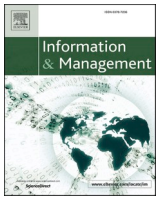
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GAI identity threat: When and why do individuals feel threatened?

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ABSTRACT

The widespread application of generative artificial intelligence (GAI) and its ability to perform various tasks have raised concerns about identity threats. This study aims to investigate antecedents, behavioral consequences, and boundary conditions of GAI identity threat. We conduct a two-phase study using a mixed-methods approach. The qualitative phase aims to identify the critical factors that influence GAI identity threat. After coding 30 semi-structured interview transcripts, we find that GAI's high intelligence and human-like abilities facilitate three kinds of affordance that influence GAI identity threat: creative affordance, analysis affordance, and communication affordance. The quantitative phase, based on IT identity threat theory, constructs a research model to examine how the three types of affordance influence GAI identity threat, as well as identifying the behavioral consequences. An examination of survey data from 405 users indicates that GAI's creative affordance, analysis affordance, and communication affordance are associated with GAI identity threat, which may lead to users' resistance behavior. Furthermore, GAI autonomy and user self-identity moderate the relationship between GAI affordances and GAI identity threat, respectively. This study identifies users' negative psychological mechanisms when using AI, develops IT identity threat theory in the context of GAI, and provides implications for practitioners to reduce identity threats.

1. Introduction

The rapid development of generative artificial intelligence (GAI) and its accelerated penetration into various realms of society have elicited new opportunities for economic and social development, as well as new challenges for humans [1]. In November 2022, OpenAI, an AI research lab, launched an intelligent conversational agent program called ChatGPT (Chat Generate Pre-Trained Transformers). Within a week after its debut, it registered one million users. Two months after its launch, it surpassed 100 million active users, becoming the fastest-growing consumer app in history. Despite its rapid growth, GAI's value has been debated, eliciting differing views [2,3]. Proponents have argued that GAI's superior processing speed and self-learning competency can improve business processes greatly [1,4]. However, opponents have taken to social networks to promote GAI threat theory and resist using it. The open letter "Pause Giant AI Experiments" signed by over 1000 people, including Elon Musk and Turing Award recipient Yoshua Bengio, has now garnered more than 9000 signatures. In addition to industry leaders, the European Union and the White House also have expressed resistance to GAI. With its human-like capabilities, GAI has the potential to replace human labor, threaten human dignity [5], and trigger heated discussions.

In the field of information systems (IS), the literature has indicated that technology can pose identity threats to humans [6]. The concept of technology identity threat is defined as "the harm caused to an individual's self-confidence due to the use of technology" [6]. This study reviews the literature on technology identity threat, revealing that existing research has focused primarily on examining antecedents of identity threat and users' attitudes and behavior toward perceived identity threats. Factors influencing identity threats include the technology's autonomy [7], technology's controllability [8], technology's anthropomorphism [9,10], perceived benefits [11], loss of status [12], threats to job security [6], loss of professional knowledge [7], and job changes [6]. Identity threats influence users' attitudes and behavior, such as experiencing negative emotions [9], willingness to engage in interactions [13], user resistance [7], and intention to continue using [11]. Scholars have examined identity threats in different technological contexts, such as smartphones [14], machines [13], social robots [9,15], and decision-support systems [11]. With the advance of technology, users also face identity threats in the context of GAI. With its powerful

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capabilities, GAI has the potential to replace humans to perform tasks such as writing, drawing, and coding. For example, AlphaCode beat about half of humans in code competitions [16]. It can solve simple computer-science problems and replace some programmers' work functions [17]. AI-generated content may even surpass human-generated content in terms of professionalism, leading to concerns that GAI may threaten people's identities [5].

Drawing on affordance theory, we explore the antecedents of GAI identity threats. Affordance theory emphasizes the possibilities for action offered to organisms by objects or technologies in the environment [18]. Applying this concept to the study of GAI can help us gain a deeper understanding of the characteristics of technology and how they affect user perception and behavior [19]. Studies have been conducted to explore technology affordance in several scenarios, such as online role-playing game [20], live streaming e-commerce [19], social networking sites [21], chatbots [22], and online learning [23]. GAI has a powerful capability that provides users with a new and wide range of action possibilities [24]. This capability makes GAI different from the traditional technologies mentioned above. Therefore, to accurately capture and analyze the reasons that users perceive GAI identity threats, we need to pay special attention to the unique affordance provided by GAI.

Through literature review, we found three research gaps. First, while existing research has examined antecedents of IT identity threat, it has failed to capture GAI's unique characteristics and the critical factors that influence users' GAI identity threat. Second, existing studies have noted that GAI may have negative effects on individuals [25,26], including identity threat [5]. However, existing research on this topic primarily has been conceptual, that is, lacking empirical research on identity threats caused by GAI and failing to examine antecedents and consequences of GAI identity threat. Third, although GAI is a powerful AI tool, its impact on different identity groups varies, and research on boundary conditions of GAI identity threat is lacking. Specifically, under what circumstances are users more likely to perceive GAI identity threat? These problems remain underexamined.

Due to the practical significance of and literature gap on GAI identity threat, we posited two research questions:

(1) Why do users perceive GAI identity threat?
(2) What are the consequences and boundary conditions of GAI identity threat?

To answer these questions, we conducted a two-phase study. During the first phase, we conducted a qualitative study to identify the critical factors that may influence users' GAI identity threat. We conducted semi-structured interviews with 30 users, and then analyzed the interview transcripts using coding techniques. This qualitative study revealed that GAI's high intelligence and human-like abilities facilitate three kinds of affordance that influence GAI identity threat: creative affordance, analysis affordance, and communication affordance. During the second phase, we conducted a quantitative study to examine how the three types of affordance influence GAI identity threat, as well as identify behavioral consequences. Based on IT identity threat theory, we constructed a research model. The empirical results indicate that GAI affordances are associated with GAI identity threat, which may lead to users' resistance behavior. Furthermore, GAI autonomy and user self-identity moderate the relationship between GAI affordances and GAI identity threat.

This study makes the following contributions: First, it identifies the critical factors that may influence users' GAI identity threat, making an original theoretical contribution. Second, it extends IT identity threat theory. Based on IT identity threat theory, it applies a research model and validates the antecedents, consequences, and boundary conditions of users' perceived identity threats in the context of GAI. Finally, it enriches the literature on technology resistance, validating that GAI identity threat may lead to users' technological resistance. It also identifies the moderating effects of GAI autonomy and user self-identity, providing insights for organizations to improve GAI design.

2. Theoretical background

2.1. IT identity threat theory

IT identity threat refers to the expected harm to people's self-beliefs due to IT use [6]. The identity threat was first proposed by Petriglieri [27]. In the workplace, people experience many things that can threaten their identity. When experiences contradict individuals' identities, the individuals' self-esteem may be threatened and they may act to maintain the self-esteem associated with their identities, which is the essence of identity threats. With the advance of technology, IT has become an important source of identity threats [6]. Scholars have explored IT identity threat in different contexts [11,28]. Craig proposed IT identity threat theory [6]. IT identity threat is triggered when individuals using IT anticipate harm to their identities, as reflected in changes in self-esteem: loss of worth-based self-esteem, loss of competence-based self-esteem, and loss of authenticity-based self-esteem [6].

According to IT identity threat theory, individuals typically have three types of identities: social, role, and personal. Social identity refers to an individual's sense of belonging to society, including the group, culture, or social network to which they belong. Role identity relates to an individual's roles and responsibilities in a particular environment or situation, such as being a family member or having an occupational position. Personal identity is an individual's unique characteristics, values, beliefs, and emotions that compose their unique self. Individuals are exposed to IT threats to their specific identities – whether social, role, or personal [6]. First, when IT reduces the power and prestige of a group, it may threaten the social identities of individuals in that group. Second, when enterprises introduce IT, individuals' job responsibilities may change. Their role identities may not be recognized adequately, leading them to feel threatened. Finally, when the use of IT is inconsistent with authentic self-beliefs, it may threaten personal identity.

Existing literature has researched the antecedents and consequences of IT identity threat. The antecedents of IT identity threat include technological autonomy [7], technological anthropomorphism [9,10], perceived benefits [11], loss of status [12], threats to job security [6], loss of professional knowledge [7], and job changes [6]. When users perceive identity threats, they can influence their attitudes and behavior, including negative emotions [9], willingness to engage [13], user resistance [7], and intention to continue using [11].

2.2. GAI identity threat

Based on the definition of IT identity threat [6], this study defines GAI identity threat as harm to individuals' self-beliefs using GAI. When individuals anticipate that the use of GAI will cause harm to their identities, GAI identity threat is triggered. Recent literature on GAI has examined its applications in various industries, such as healthcare [26], education [29], marketing [30], and finance [4]. GAI with human-like competencies can perform various complex tasks, including translating, writing, and coding [2,31]. GAI is superior to humans in some respects and has the potential to replace some human roles in specific situations, triggering both excitement and fear about GAI [5]. GAI has characteristics that traditional technology does not have [32,33]. Therefore, IT identity threat theory in new scenarios needs to be developed continuously.

Compared with IT identity threat in the traditional human–computer interaction context, identity threats among individuals in the new GAI context have three main unique characteristics: First, GAI identity threat has different antecedents from traditional IT identity threat. Previously, IT did not have the human-like interaction competency of GAI [2]. When GAI can create content [24], make decisions [33], and express emotions like humans [34], it may lead people to feel that their identities are

threatened. Second, the impact of GAI identity threat is more far-reaching. In the past, IT identity threat research was concerned about machines replacing human jobs [7], but with the development of technology, people began to worry that GAI was threatening human dignity and replacing human intelligence and competency [4]. Finally, the audience for GAI identity threat is much broader. Previously, the audience for IT identity threat may have been IT professionals, but due to the powerful capabilities of GAI, the audience for GAI identity threat is much broader and may include creative workers, such as painters and poets. Even "safe" professions such as teachers, academics, programmers, and analysts may be at risk of being replaced by GAI. However, no empirical research has been conducted on GAI identity threat.

2.3. GAI affordance

Affordance has its origins in ecopsychology and refers to the possibilities for action that objects and spaces in the environment offer to organisms [18]. The theory has been extensively researched in the field of IS and has been applied in product design and technology development to explain how products or technologies affect people's perceptions and actions [19], as shown in Table 1. Scholars have explored affordance in different scenarios, focusing on the functions of technology [35]. Lee et al. [22] applied the theory to identify chatbot affordances (anytime anyplace connectivity, information association, visibility, interactivity), highlighting their impact on perceived competence and perceived warmth. Furthermore, Zhan et al. [36] define interactivity availability, agent availability, and security availability as AI affordances, exploring the impact of AI affordances on users' perceived fears. However, these definitions and dimensions of affordance are still based on traditional AI technology contexts and do not reflect the unique characteristics of GAI.

Based on existing research, we define GAI affordances as the possibilities of action that GAI provides to the user. Compared with traditional AI (which only provides decision support to customers), GAI has the following features: First, GAI can generate new content based on user input [1,2]. This makes GAI uniquely suited for certain domains, such as artistic and literary creation [24,32]. In contrast, traditional AI focuses more on simulation and prediction. Second, GAI can generate logical natural language text based on user prompts and interact smoothly with users [34]. Traditional AI often lacks flexibility and diversity in generating natural language text, making it difficult to adapt to different user needs and scenarios. Third, GAI adopts more advanced machine learning techniques that can excel in handling complex multimodal data and provide valuable information [30]. Traditional AI usually relies on manually defined rules, making it difficult for it to handle complex and variable data. Therefore, further research is needed to investigate affordances in the context of GAI.

Previous studies have found that technology affordance affects people's perceptions and behavior, such as cyberbullying [21], technological fear [36], technological addiction [20], cognitive load [19], and brand loyalty [22]. Due to its powerful features and wide application potential, GAI has the potential to replace humans in technical tasks such as writing, drawing, and coding [17], resulting in individuals feeling that their identities are threatened. Therefore, this study explores users' perceived GAI identity threats from an affordance perspective.

3. Research design

We employed a mixed-methods approach that comprised qualitative and quantitative stages. Using mixed methods is appropriate. First, mixed methods are suitable for answering both validation and exploratory research questions [38]. Second, they are an effective way to develop new theoretical perspectives [39]. The mixed methods approach can help researchers understand a research problem comprehensively, by providing an in-depth understanding of the phenomenon (qualitative) and by testing causal relationships based on a large number of quantitative data (quantitative). The mixed-methods approach helps increase the realism and rigor of scientific research.

In terms of generative AI, the potential antecedents of user-perceived GAI identity threat have not been examined in depth. Moreover, the literature on consequences and boundary conditions of user-perceived identity threats in this new context is lacking. Therefore, we conducted a qualitative study (Study 1) to examine the antecedents of user-perceived GAI identity threat. Then we applied the findings to a quantitative study (Study 2). Based on the antecedents of GAI identity threat identified in Study 1 and IT identity threat theory, Study 2 developed a theoretical model to validate the antecedents, consequences, and boundary conditions of GAI identity threat. The method roadmap is presented in Fig. 1.

Table 1
Summarizing the application of affordance theory in the IS field.

References	Research question	Influencing factors	Outcome variables	Methodology	Research context
(Volkoff & Strong 2013) [35]	How do affordances arise from the relationship between organizations and IT artifacts?	Basic affordances, standardizing affordance, communication affordance, nalysis affordance	NA	Case study	IT-associated organization changes
(Chan et al., 2019)[21]	How do SNS affordances affect the assessment of SNS environmental conditions?	Accessibility affordance, information retrieval affordance, editability affordance, and association affordance	Social networking sites bullying	Survey	Social networking sites bullying
(Lee et al., 2021)[20]	How are MMORPG affordances associated with the duality of hedonic effects and the extent of MMORPG addiction?	Achievement affordance, social affordance, immersion affordance	Massively multiplayer online role-playing game addiction	Survey	Massively multiplayer online role-playing game addiction
(Lee et al., 2023)[22]	How do chatbot affordances influence customer perceptions and behavioral intentions?	Anytime anyplace connectivity, information association, visibility, interactivity	Perceived chatbot competence, perceived chatbot warmth	Survey	Brand loyalty
(Zhang et al., 2023)[23]	How do technology affordances influence user experience in AEILS?	Interactivity affordance, personalization affordance, competition affordance, convenience affordance	Continuous intention, speaking anxiety	Survey	AI-enabled intelligent learning system (AEILS)
(Zhan et al. 2023)[36]	How do perceived bandwidth, contingency, and synchronicity influence the different dimensions of fear of AI?	Interactivity affordances, agency affordances, security affordances	Perceived fear	Survey	Fear of AI technology
(John et al., 2024)[37]	How do team-based gamification affordances affect coping responses?	Collaboration affordance, competition affordance	Task-focused coping, emotion-focused coping, avoidance coping	Survey	Enterprise resource planning
(Hua et al., 2024)[19]	How does live streaming influence customer purchase intent?	Visibility, interactivity, entertainment, broadcasting, source diversity	Vicarious learning, cognitive load	Mixed methods	Live streaming e-commerce

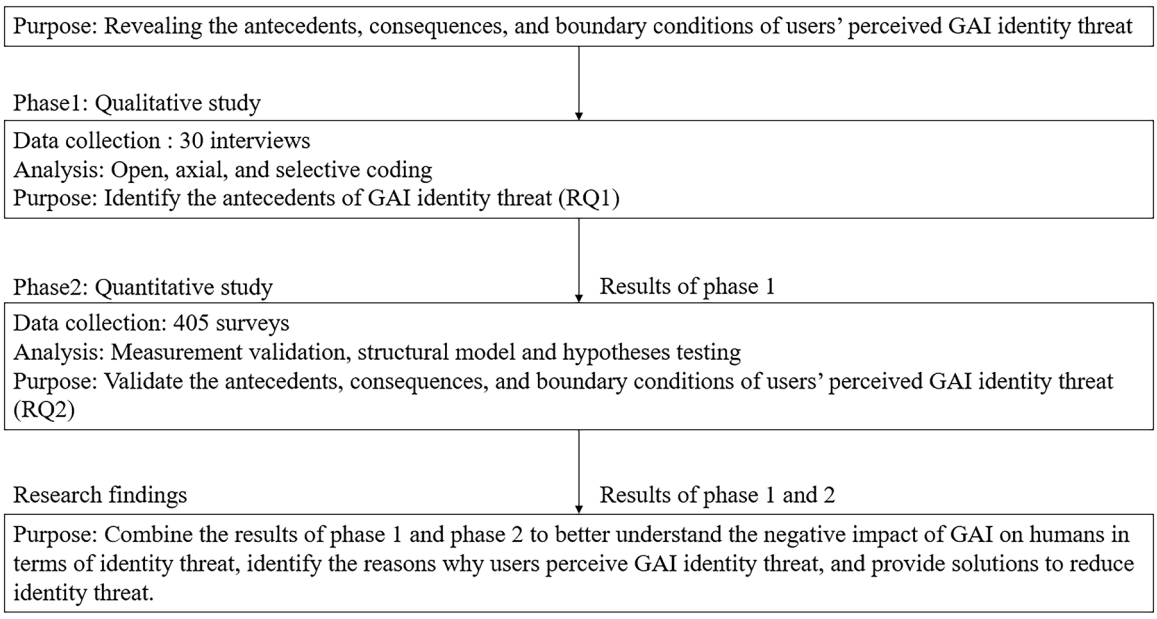


Fig. 1. The method roadmap.

4. Study 1: qualitative research

4.1. Data collection

During the first phase, we conducted a qualitative survey to identify the critical factors that may influence users' GAI identity threat. In April 2024, we interviewed 30 people (16 male and 14 female) who had experience using GAI. During the interviews, we asked the respondents about (1) their overall experiences with generative AI; (2) generative AI's primary functions and applications in practical scenarios; and (3) the reasons that people perceive threats when using generative AI. In selecting our sample, we looked on social media platforms for respondents with experience using generative AI, and then encouraged respondents with experience using generative AI to invite people with similar experiences to participate. This approach was effective in reducing self-selection bias. For each interview, we provided a brief introduction to the research context and rewarded each interviewee with 40 RMB (approximately 6 US$). All interviews were open-ended, all personal information was anonymized, and each interview lasted approximately 30 min.

4.2. Coding process

We analyzed the interview documents through coding. Two authors conducted the coding process to ensure the reliability and validity of the results. The process comprised three phases. During the first phase, we reviewed and summarized the existing literature on topics related to GAI and identity threats to establish a theoretical framework. To ensure the reliability of coding results, we randomly selected 10 % of the sample for double-blind independent pre-coding. Two coders compared and discussed the pre-coding results and then modified the coding scheme based on the results. During the second phase, we conducted open coding. Two coders independently and repeatedly read the transcripts and annotated the text to identify all possible antecedents of GAI identity threat. The coders marked all discrete concepts and phrases related to the research questions for further analysis. During the third phase, the two coders independently conducted axial coding based on the open coding results. We then invited experts to check and analyze the codes until a consensus was reached.

4.3. Coding results

After coding the 30 semi-structured interview materials, we found that GAI's high intelligence and human-like abilities facilitated three kinds of affordances that influenced GAI identity threat: creative affordance, analysis affordance, and communication affordance. The

Table 2
Examples of text data, open codes, and axial codes.

Axial codes	Open codes	Text data example
Creative affordance	Create pictures, Create according to the prompt, Create text	A1. "AI creates pictures that can confuse judges at certain painting exhibitions, possessing the ability to paint like human beings." A2. "We just need to give him the prompt and the AI will autonomously create according to the prompt." A3. "The characteristic of GPT is its ability to quickly generate a large amount of text."
Analysis affordance	Quick analysis, Clear analysis, Analyze problems	B1. "It can quickly analyze the question and give me the answer I want." B2. "It will help me to expand my thoughts and clearly analyze the overall logic of the material." B3. "ChatGPT can help me analyze the problems with the code and help me fix the bugs in the code."
Communication affordance	Human-like communication, Emotional talking, Coherent conversations	C1. "I think sometimes it replies with a human tone, which gives me a feeling of wanting to communicate with it as a human being, it's quite polite." C2. "Some people would use it as an emotional assistant and would talk to him." C3. "It is capable of conducting coherent conversations."

coding results are shown in Table 2. To ensure the validity of the study, all the factors listed in this study were mentioned multiple times by different respondents, as shown in Table 3.

Creative affordance refers to GAI's ability to generate new ideas and discoveries in a given field. Creative affordance is the most frequently mentioned antecedent of GAI identity threat. Previous research has pointed out that robots with the ability to innovate can threaten the status of people [13]. From the interview texts, we found that users perceive GAI to be highly creative. GAI can generate new content such as text, images, and video based on prompts that users enter.

Analysis affordance refers to GAI's ability to choose effectively and achieve a purpose when facing problems based on already accumulated experience. Analysis affordance is another critical antecedent of GAI identity threat. GAI can translate and code, reducing some repetitive work that threatens users' professional identities and economic status [17]. From the interview texts, we found that users perceived GAI as having powerful analytical capabilities. Users can use GAI for analysis to find solutions.

Communication affordance refers to GAI's ability to understand the user's feelings and respond reasonably to actual scenarios. Communication affordance is the third critical antecedent of GAI identity threat. Previous studies have pointed out that intelligent customer service with communication affordance will replace part of the work of manual customer service [40]. When people interact with GAI, it not only understands the feelings and emotions behind the language but also generates emotional responses based on user inputs and contexts. This communication affordance makes GAI's dialogue with people more natural. From the interview texts, we found that users perceived GAI as having human-like communication skills and being able to hold coherent conversations.

5. Study 2: quantitative research

5.1. Hypothesis development

During the second phase, we conducted an empirical study to verify the antecedents, consequences, and boundary conditions of GAI identity threat. Based on IT identity threat theory, we constructed a research model to examine how the three types of affordance influence GAI identity threat and identity the behavioral consequences. Furthermore, by validating the moderating roles of GAI autonomy and user self-identity, we determined which user groups are more vulnerable to GAI identity threat and provided solutions to reduce identity threats. The research model is presented in Fig. 2.

5.1.1. GAI affordances and GAI identity threat

Creative affordance refers to GAI's ability to generate new ideas and discoveries in a given field. Previous research has indicated that robots with the ability to innovate can threaten the status of people [13]. GAI possesses remarkable creative capabilities, as it can generate new content such as text, images, and audio after learning large amounts of data [24]. This creative affordance allows GAI to offer unique advantages, bringing a new creative experience to humans. Extant research indicates that the creativity of AI in the arts poses an ontological threat to the anthropocentric worldview [41]. Previously, creativity was viewed as a unique human characteristic, but the emergence of GAI has broken the boundaries of creativity and raised concerns about identity threats. With superior creativity, GAI can automatically generate novels, poems, images, and musical scores, partly replacing human work [42]. This not only will cause existential problems, such as unemployment, but also will erode humans' sense of ontological security about their unique position in the world. Threats to unique status are critical factors constituting personal identity threats [6]. Therefore, we hypothesized the following:

Table 3
Generative AI affordances for users.

Generative AI affordances	Times mentioned	# of People
Creative affordance	39	24
Analysis affordance	33	21
Communication affordance	22	14

Hypothesis 1 (H1). *GAI's creative affordance is positively associated with GAI identity threat.*

Analysis affordance refers to GAI's ability to choose and achieve purposes effectively when facing problems based on already accumulated experience [43]. GAI can identify users' needs and provide actionable suggestions, which may lead to automation of some tasks, such as translation and coding, thereby threatening users' professional identity and economic status [5]. For example, AlphaCode beats about half of humans at code competitions [16]. It can solve simple computer-science problems and replace some work of programmers [17]. Students can interact with ChatGPT and query the appropriate knowledge, replacing part of the teacher's work in helping students solve problems [44]. The scope of many professions' work has been impacted and has become less irreplaceable due to GAI's powerful analysis affordance [1]. GAI's analysis capabilities are gradually changing the social status and job scopes of certain professions. People's social status and the role identities derived from their work are important components of personal identity [6]. Therefore, we hypothesized the following:

Hypothesis 2 (H2). *GAI's analysis affordance is positively associated with GAI identity threat.*

Communication affordance refers to GAI's ability to understand users' feelings and respond reasonably to actual scenarios [45]. Previous studies have noted that intelligent customer service with communication affordance will replace part of the work of manual customer service employees [40]. Communication affordance is an indispensable and important affordance for machines to realize intelligence. When people interact with GAI, based on natural language processing technology, GAI can provide professional and friendly support to generate pleasant dialogue [34]. For example, when you share a happy story with ChatGPT, it may respond with a smiley face emoticon, conveying a sense of affection and understanding [46]. This communication affordance makes the dialogue between GAI and people more natural and closer to human-to-human communication. However, people may feel that the machine is too intelligent and fear that it will threaten them [5,37]. When GAI demonstrates a high level of communicative intelligence, including the ability to mimic human emotion comprehension and expression, individuals may feel inadequate, which can damage self-esteem. Loss of self-esteem is a significant factor in identity threats [6]. Simultaneously, GAI with communication intelligence may replace part of the work of human customer service employees. Therefore, we hypothesized the following:

Hypothesis 3 (H3). *GAI's communication affordance is positively associated with GAI identity threat.*

5.1.2. GAI identity threat and users' resistance behavior

GAI identity threat refers to the harm caused to an individual's self-beliefs by using GAI. Currently, many users are expressing their concerns about GAI on social media platforms. With continuous advances in AI technology, GAI will be able to perform tasks that humans cannot, resulting in the loss of human competitiveness and survivability [2]. Thus, the scope of work in many professions becomes less irreplaceable [24]. People may feel anxiety when interacting with generative AI, feeling that the machine is too intelligent and worrying that it threatens them [1,5]. According to IT identity threat theory, identity threats elicit responses [6]. Out of fear and suspicion over threats, people experience psychological and behavioral changes in response to these threats, such

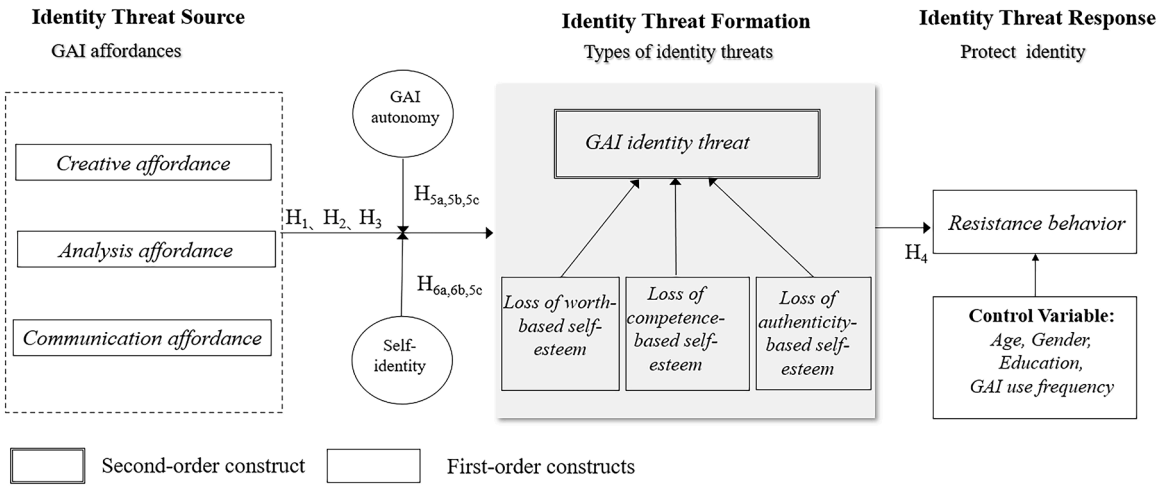


Fig. 2. The research model.

as self-defense [47], resistance to use [48], refusal to engage with technology [49], formation of an anti-identity [50], compensatory consumption [51], and negative attitudes toward bots [52]. GAI is currently generating intense discussions on the Internet, and identity threats that GAI poses to users may trigger resistance behavior. Therefore, we hypothesized the following:

Hypothesis 4 (H4). *GAI identity threat is positively associated with resistance behavior.*

5.1.3. The moderating effects of GAI autonomy

Autonomy refers to the ability of systems to perform human tasks without specific human intervention [53,54]. Previous research has noted that autonomous robots threaten human identity, uniqueness, safety, and resources [7,55]. In the workplace, autonomous service robots may increase the insecurity of hotel employees, as they may be at risk of being completely or partially replaced by service robots [56]. GAI autonomy refers to GAI's ability to generate content autonomously without human intervention. Anxiety and insecurity can be triggered when people interact with AI that seems to be able to "think" and "act" autonomously, particularly if they feel that this autonomous entity can replace them or outperform them in some tasks [57]. When users think that they cannot control the AI system, the sense that this intelligence is a threat to their identity increases, as they view the system as an entity with autonomous intentions, rather than a tool that operates under their command [7]. GAI autonomy may enhance the relationship between GAI affordances and GAI identity threat. Therefore, we hypothesized the following:

Hypothesis 5 (H5). *GAI autonomy will positively moderate the relationship between GAI affordances (creative affordance [H5a], analysis affordance [H5b], and communication affordance[H5c]) and GAI identity threat.*

5.1.4. The moderating effects of individuals' self-identity

Self-identity refers to individuals' cognition and evaluation of their identities, values, competencies, and social status, as well as their emotional responses to these cognitions and evaluations [58]. Individuals pay different roles in society. When individuals experience changes in their environment, a stable self-identity becomes a solid backbone for coping with these changes [59,60]. Individuals with high social status generally receive more recognition and have a more prominent self-identity than those with low status [61]. GAI, with its superior capabilities, makes it easy for people without corresponding skills to attempt more challenging tasks, such as writing, painting, coding, and data analysis [2]. While previous concerns surfaced that mechanical automation would replace some manual or repetitive work, due to the high intelligence level of GAI, it is now capable of performing creative work, impacting some "high identity" occupations, such as teachers, academics, programmers, and analysts. GAI affordances expose individuals with a strong sense of self-identity to stronger identity threats. Therefore, we hypothesized the following:

Hypothesis 6 (H6). *Individuals' self-identity will positively moderate the relationship between GAI affordances (creative affordance [H6a], analysis affordance [H6b], and communication affordance[H6c]) and GAI identity threat.*

5.2. Method

5.2.1. Measurement and data collection

We used a survey method to conduct this research, which was appropriate for this study: First, the number of GAI users is increasing rapidly, generating intense discussions [62]. Therefore, qualified respondents were easy to find. Second, the variables in the research model involved antecedent and consequent processes of users' perceived identity threats, which lend themselves to the use of scales to measure users' psychological attributes. The questionnaire was designed based on scientific scales. Furthermore, to ensure that respondents had experience interacting with GAI, we used three methods to screen respondents. Method 1: We attached a "Respondent Guide" to the front of the questionnaire to remind respondents of the following: "This questionnaire is designed for users who have used GAI. If you have not used GAI, please do not answer the following questions. When answering the questions, please recall your recent experience interacting with GAI." Method 2: We designed a logical link between the questions at the beginning of the questionnaire: "Have you ever used GAI?" If the user selected "No," the subsequent questions on the questionnaire were not displayed on the electronic questionnaire. Method 3: We asked the platform to screen the respondents for us, and the people who filled out the questionnaire were the working population because students' identities were less likely to be threatened.

The scales used have been verified in previous studies to have good reliability and validity, as shown in Table A.1 in Appendix A. This study utilized a 7-point Likert scale (1 = "completely disagree" to 7 = "completely agree"). In the first part, we briefly introduced the study's purpose and then asked questions related to GAI affordances, GAI identity threat, and resistance behavior. In the second part, we asked about demographic characteristics, such as age, gender, education, and use frequency.

We conducted a small-scale pilot test with two professors and five PhD students to ensure that the questionnaire was logical, accurate, and

easy to understand. We then collected 405 valid responses from Wenjuanxing.com, a Chinese professional survey platform and one of China's largest online survey platforms, which can provide large-scale data collection. We screened the questionnaires based on the following procedures. First, we excluded questionnaires in which responses were overwhelmingly consistent across virtually all questions, indicating that the respondents may not have filled out the questionnaire carefully. Second, we excluded questionnaires that provided incorrect answers to our trap questions. Finally, we identified the remaining 405 valid questionnaires, of which 47.2 % were male and 52.8 % were female. Most respondents were between 30 and 39 years old, composing 43.0 % of the total sample. The proportion of respondents between 20 and 29 years old was 32.8 %. Regarding use frequency, 30.1 % of respondents said they used GAI multiple times a week, the highest percentage in the sample. See Table A.2 in Appendix A for demographics.

5.2.2. Measurement model test

In this study, Smart PLS 4.0, based on partial least squares (PLS), was used to analyze the reliability and validity of the measurement model and then evaluate the structural model. Compared with covariance-based structural equation modeling (CB-SEM), PLS-SEM is more suitable for exploratory studies or theory building [63]. Furthermore, the GAI identity threat in this study model is a formative structure, and the model is complex, which is more suitable to analyze with PLS.

In the measurement model examination, we conducted an exploratory factor analysis and then tested the reliability and validity of the scale. The factor loadings and cross loadings in PLS are shown in Table A.3. The loading values on the corresponding factor were higher than 0.7, while the values on other factors were lower than 0.5, indicating that the scale had good discriminant validity. Table A.4 in Appendix A presented the scale properties, with Cronbach's and composite reliability (CR) values above 0.7, indicating good reliability of the scale. The average variance extracted (AVE) values for each factor were higher than the required value of 0.5, and the standardized loadings of the items were above the 0.7 level and were all significant at 0.001, indicating that the scale had good convergent validity.

To further examine the discriminant validity, we compared the AVE values' square root and the items' correlations [64]. The results (Table A.5 in Appendix A) indicated that the square roots of the AVE values were higher than the corresponding correlation values, indicating good discriminant validity in the scale. Simultaneously, we examined the values of the heterotrait–monotrait ratio (HTMT). The results (Table A.6 in Appendix A) indicated that the value of HTMT did not exceed the recommended threshold of 0.90 [65], indicating that the scale had good discriminant validity.

To ensure the validity of the second-order variables, we examined the significance of the indicator weights. The results (Table A.7 in Appendix A) indicated that the weights of loss of worth-based self-esteem, loss of competence-based self-esteem, and loss of authenticity-based self-esteem (0.33, 0.31, and 0.36, respectively) significantly impacted GAI identity threat, indicating their importance in forming GAI identity threat. Second, we used the variance inflation factors (VIF) to check whether the first-order structures were highly correlated. The results indicated that the VIF values of these first-order structures were all below the threshold of 3.3, indicating that multicollinearity was not a problem for our measure [66].

To ensure that the dataset was free of common method bias (CMB), we adopted pre-avoidance and post-test methods. First, when collecting data, in addition to ensuring the anonymity of respondents, we set one attention trap question to help us screen out questionnaires that were not answered carefully. We then conducted a Harman one-way test. The results indicated that nine factors were extracted, with the first factor accounting for only 10.28 % of the variance, indicating that CMB was not a significant problem in this study. Finally, we examined the data using the marker variable method. We selected the item "How concerned are you about your health?" as the marker variable and compared the path coefficients of the model, incorporating this marker variable with those of the base model. The results are shown in Table A.8 in Appendix A. The marker variable exerted no significant effect on GAI identity threat or resistance behavior, with correlations of 0.02 and –0.05, respectively, close to zero. Meanwhile, the results of the marker variable model were similar to those of our base model, and the results from all assumptions remained qualitatively unchanged. Thus, CMB was not a significant issue in this study.

5.2.3. Structural model test

We used SmartPLS 4.0 to test the structural model, which explained 25.8 % of the variance in resistance behavior and 37.3 % of the variance in GAI identity threat. The results of the main effects are shown in Table 4. Creative affordance ($\beta = 0.25$, $t = 4.41$, $p < 0.001$), analysis affordance ($\beta = 0.24$, $t = 5.34$, $p < 0.001$), and communication affordance ($\beta = 0.20$, $t = 4.77$, $p < 0.001$) all had significant positive effects on user-perceived GAI identity threat, thereby supporting H1, H2, and H3. Meanwhile, user-perceived GAI identity threat had a significant positive effect on resistance behavior ($\beta = 0.82$, $t = 11.31$, $p < 0.001$), supporting Hypothesis 4. Furthermore, we tested the effect of the control variables, and the results revealed insignificant effects on the dependent variable.

After the main effect was tested, the moderating effects of GAI autonomy and user self-identity were examined. As shown in Table 5, the results indicate that GAI autonomy had a significant positive moderating effect on the relationship between GAI affordances and GAI identity threat. High GAI autonomy strengthened the positive relationship between analysis affordance and GAI identity threat. High GAI autonomy strengthened the positive relationship between communication affordance and GAI identity threat. Thus, H5b and H5c were supported. Self-identity had a significant positive moderating effect on the relationship between GAI affordances and GAI identity threat. High self-identity strengthened the positive relationship between analysis affordance and GAI identity threat. High self-identity strengthened the positive relationship between communication affordance and GAI identity threat. Thus, H6b and H6c were supported. Furthermore, the results indicated that autonomy and self-identity did not moderate the relationship between creative affordance and GAI identity threat. Therefore, H5a and H6a were not supported, which indicates that creative affordance always affects identity threats in the same way regardless of the level of autonomy and self-identity. The potential explanation is that creativity is the ultimate expression of human intelligence [41,42]. Furthermore, with the continuous advance of technology, GAI can autonomously create text, images, and audio. People cannot even distinguish between AI-generated works and human-generated works [42]. This may be the reason that, unlike the other two types of GAI affordance, the impact of creative affordance on GAI identity threat is not moderated by GAI autonomy and self-identity.

5.2.4. Mediating effect

To examine whether GAI identity threat mediated the relationships

Table 4
Results of the main effect.

Hypothesis paths	Path coefficients	*P*-value	Supported or not
H1: Creative affordance → GAI identity threat	0.25***	<0.001	Supported
H2: Analysis affordance → GAI identity threat	0.24***	<0.001	Supported
H3: Communication affordance → GAI identity threat	0.20***	<0.001	Supported
H4: GAI identity threat → Resistance behavior	0.82***	<0.001	Supported

*** $p<0.001$;
** $p<0.01$; * $p<0.05$.

Table 5
Results of the moderating effect.

Hypothesis paths	Path coefficient	*P*-value	Supported or not
H5a: The moderating effect of GAI autonomy on H1	−0.04	=0.214	Unsupported
H5b: The moderating effect of GAI autonomy on H2	0.10**	=0.006	Supported
H5c: The moderating effect of GAI autonomy on H3	0.09**	=0.004	Supported
H6a: The moderating effect of self-identity on H1	−0.07	=0.129	Unsupported
H6b: The moderating effect of self-identity on H2	0.08*	=0.011	Supported
H6c: The moderating effect of self-identity on H3	0.10***	<0.001	Supported

*** $p < 0.001$;
** $p < 0.01$;.
* $p < 0.05$.

between GAI affordances and resistance behavior, we assessed the mediating effect based on a bootstrapping method with a resampling size of 5000. As shown in Table 6, the indirect effects of creative affordance, analysis affordance, and communication affordance on resistance behavior were significant. The direct effect of creative affordance on resistance behavior was also significant, while the direct effects of analysis affordance and communication affordance on resistance behavior were not significant. Thus, creative affordance enhanced resistance behavior, partially mediated by GAI identity threat. Meanwhile, analysis affordance and communication affordance enhanced resistance behavior under the fully mediating effect of GAI identity threat.

6. Discussion

6.1. Discussion of the results

This study comprised two phases using a mixed-methods approach to investigate the antecedents, behavioral consequences, and boundary conditions of GAI identity threat. The first phase, a qualitative study, aimed to identify the critical factors that may influence users' GAI identity threat. After coding the 30 semi-structured interview transcripts, we found that GAI's high intelligence and human-like abilities facilitated three kinds of affordances that influence GAI identity threat: creative affordance, analysis affordance, and communication affordance. The second phase, a quantitative study, constructed a research model to examine how the three types of affordances influence GAI identity threat. We validated the model using survey data from 405 users. The results indicated that GAI's creative affordance, analysis affordance, and communication affordance significantly influence users' perceived identity threat, which may lead to users' resistance behavior. Furthermore, GAI autonomy and user self-identity moderate the relationship between GAI affordances and GAI identity threat, respectively.

In research on IT identity threat, scholars have examined identity threats in different technological contexts, such as smartphones [14], machines [13], social robots [9], and decision-support systems [11]. The antecedents of IT identity threat include technological anthropomorphism [9], technological autonomy [7], and loss of status [12]. With the development of technology, users also face identity threats in the context of GAI. With powerful capabilities, GAI has the potential to replace humans in tasks such as writing, drawing, and coding. This study investigated user identity threats in the context of GAI, identifying critical factors that may influence users' GAI identity threat and enriching IT identity threat literature. Extant studies on GAI identity threat have primarily provided conceptual descriptions of how GAI might pose a threat to personal identity [5,24]. In this study, we investigated the antecedents, intrinsic mechanisms, and consequences of user identity threats in the context of GAI empirically through surveys. Our research model suggests that GAI service providers should be aware that GAI may have a negative impact on people. The impact of GAI varies across identity groups. In this study, we examined when users perceive that GAI poses more serious identity threats and proposed solutions to reduce threats.

6.2. Theoretical implications

First, the findings provide a comprehensive understanding of technology affordance and enrich the affordance literature based on the GAI context. Although existing studies have explored AI affordance in different scenarios [22,23], they have not captured the unique characteristics of GAI or explored user-perceived GAI identity threats from the perspective of GAI affordance. This study adopts a qualitative research methodology to explore the key factors influencing users' GAI identity threats. After coding 30 semi-structured interview materials, we identified three types of GAI affordance, which influence users' GAI identity threat: creative affordance, communicative affordance, and analysis affordance. In doing so, we make an original theoretical contribution by providing a more comprehensive explanation of technological affordance in the GAI context, as well as a classification basis for future GAI affordances.

Second, this study extends IT identity threat theory in the context of GAI. Scholars have examined identity threats in different technological contexts, such as smartphones [14], machines [13], social bots [15], and decision-support systems [11]. However, with the continuous development of technology, GAI such as ChatGPT continues to be applied to people's work and daily life and has changed the business processes of organizations. Currently, IT identity threat theory has not been applied in the context of GAI. This study extends the application scenarios of IT identity threat theory. Based on IT identity threat theory, this paper comprehensively verifies the antecedents, consequences, and boundary conditions of user-perceived identity threats in the context of GAI. The antecedents of IT identity threat are technological anthropomorphism [9], technological autonomy [7], and loss of status [12]. In this study, we captured GAI affordances (creative affordance, analysis affordance, and communication affordance) as threat sources, validated users' resistance behaviors after perceiving threats, and enriched the IT identity threat literature.

Third, this study enriches the technological resistance literature by demonstrating that GAI identity threat may lead to users' technological resistance. This study also identifies the moderating effect of GAI autonomy and user self-identity. Technological resistance is the most

Table 6
Results of the mediating effect.

Independent variable	Mediating variable	Dependent variable	Direct effect			Indirect effect			Results
			Effect	LLCI	ULCI	Effect	LLCI	ULCI	
CRA	GIT	RB	−0.17	−0.31	−0.03	0.35	0.25	0.47	Partial
ANA	GIT	RB	−0.09	−0.20	0.03	0.30	0.20	0.40	Full
COA	GIT	RB	0.03	−0.08	0.13	0.23	0.15	0.32	Full

Note: Level of confidence = 95 %; LLCI/ULCI = lower/upper limit of the confidence interval; CRA = creative affordance; ANA = analysis affordance; COA= communication affordance; GIT = GAI identity threat; RB= resistance behavior.

important problem that needs to be solved in the development of AI. In the psychology and management literature, some scholars have proposed solutions such as user training, participation in information sharing, and rewarding business improvement. However, these solutions are not applicable in the context of GAI. In this paper, we propose that reducing GAI autonomy can significantly weaken the direct relationship between GAI affordances and GAI identity threat, thereby reducing users' resistance to GAI. Meanwhile, we accurately capture the population of perceived identity threats. The findings reveal that not all users feel the same identity threats and that high self-identity may enhance the effect of GAI affordances on identity threats. We capture the characteristics of the population that perceives identity threats, and further research can be conducted on specific populations.

6.3. Practical implications

This study also poses some implications for practice. First, our findings suggest that GAI service providers should be aware that GAI may have a negative impact on humans. GAI can generate content autonomously based on users' needs, replacing human experts in providing users with solutions to problems. Sometimes, people cannot distinguish between AI-generated content (AIGC) and professionally generated content (PGC). Therefore, perceived identity threats are an important issue when users receive GAI services. Bots with high intelligence can pose severe threats to humans, inspiring negative emotions in users. GAI service providers looking to expand their user base should consider identity threats to users to make the technology beneficial to humans.

Second, GAI service providers should consider the moderating effect of GAI autonomy. Our study reveals that GAI autonomy moderates the relationship between GAI affordances and identity threats. Users tend to worry that GAI is too autonomous to be controlled by humans. Users fear that GAI will replace them. Increasing controllability can build people's trust in technology and reduce their perceived identity threats, which in turn reduces users' resistance behavior. Therefore, GAI service providers should increase the control features of GAI, providing users with customizable parameters, such as response length, speed, and tone style.

Third, GAI service providers should understand that users with high self-identity are more sensitive to GAI identity threat. Our study finds that self-identify moderates the relationship between GAI affordances and GAI identity threat. People with high levels of self-identity value their identities and tend to have some social status as well. In the past, concerns surfaced that mechanical automation would replace some manual or repetitive jobs, but due to GAI's high intelligence level, it can engage in some creative work, thereby influencing some "safe" occupations. Therefore, to reduce perceived identity threats, GAI service providers should pay more attention to people with high self-identity.

6.4. Limitations and future research

This study has several limitations and offers directions for future research. First, it examines the largely underexamined topic of GAI identity threat. All results were obtained in China; the results and conclusions do not take cultural differences into account. Future research should incorporate cross-cultural factors and use global data for further validation. Second, this paper investigates GAI identity threat, but different types of GAI may vary in design approach and may pose different levels of identity threats to users. Future research could examine the threat posed by certain types of GAI. Third, our research focuses only on GAI identity threat, which may mean that other potential negative factors are overlooked, and future research should examine other possible antecedents that may lead to users refusing to use it, such as bias and lack of transparency.

7. Conclusions

We conduct a two-phase study using a mixed-methods approach to investigate the antecedents, consequences, and boundary conditions of GAI identity threat. The first phase, a qualitative study, aims to identify the critical factors that influence GAI identity threat. After coding the 30 semi-structured interview transcripts, we find that GAI's high intelligence and human-like abilities facilitate three kinds of affordance that influence GAI identity threat: creative affordance, analysis affordance, and communication affordance. The second phase, a quantitative study, constructs a research model to examine how the three types of affordance influence GAI identity threat, as well as the behavioral consequences. The examination of survey data from 405 users indicates that GAI affordances are positively associated with GAI identity threat, which may lead to users' resistance behavior. Furthermore, by validating the moderating effects of GAI autonomy and user self-identity, we identify which user groups are more vulnerable to GAI identity threat and provide solutions to reduce identity threats. The results indicate that GAI autonomy and user self-identity moderate the relationship between GAI affordances and GAI identity threat. This study enriches the identity threat literature in the context of GAI and provides theoretical guidance for organizations to promote technology that better serves humanity.

CRediT authorship contribution statement

Jing Zhou: Writing – original draft, Validation, Software, Methodology, Formal analysis. **Yaobin Lu:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Qian Chen:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1, Table A2, Table A3, Table A4, Table A5, Table A6, Table A7, Table A8

Table A.1
The measurement scale.

Constructs	Items
Creative affordance (Adapted from [67,68])	1. This GAI can generate a lot of unique ideas. 2. This GAI can generate a lot of novel ideas. 3. This GAI often provides creative solutions to the problems we raise.
Analysis affordance (Adapted from [43])	1. This GAI can collaborate with us in some analytical work. 2. This GAI can help me perform some analytical work. 3. This GAI can provide a comprehensive analysis of the issues I have raised.
Communication affordance (Adapted from [69])	1. This GAI has good communication skills. 2. This GAI can interact with me effectively. 3. I find it difficult to interact with this GAI. (Reverse coded)
Loss of worth-based self-esteem (Adapted from [6])	1. This GAI can replace my job, making me feel less respected among my peers. 2. This GAI can replace my job, making me feel my peers will lose respect for me. 3. This GAI can replace my job, making me feel others will think I am a poor member of my peer group. 4. This GAI can replace my job, making me feel that my peers will have less admiration for me.
Loss of competence-based self-esteem (Adapted from [6])	1. This GAI can replace my job, making me feel less confident that I can do my job well. 2. This GAI can replace my job, making me feel less confident about having the skills needed to get work done. 3. This GAI can replace my job, making me feel that I have less ability to get work done. 4. This GAI can replace my job, making me feel that I do things poorly."
Loss of authenticity-based self-esteem (Adapted from [6])	1. This GAI can replace my job, making me feel displeased with who I am. 2. This GAI can replace my job, making me feel less like the person I want to be. 3. This GAI can replace my job, making me discouraged with who I am. 4. This GAI can replace my job, making me take a less positive attitude toward myself as a person.
Resistance behavior (Adapted from [70] [71])	1. I prefer to seek human help rather than use this GAI. 2. I will try to avoid using the information provided by this GAI. 3. I do not agree with the suggestions provided by this GAI.
Autonomy (Adapted from [53])	1. This GAI can autonomously generate content for me. 2. This GAI can autonomously provide me with information. 3. This GAI has the ability to generate content autonomously.
Self-identity (Adapted from [72,73])	1. I have a strong sense of belonging to the group represented by my identity. 2. I have a strong affinity for my own identity. 3. My identity is like a part of myself. 4. My identity holds deep personal significance for me.

Table A.2
Demographics of the research sample (n = 405).

Measure	Item	Frequency	Percentage (%)
Gender	Male	191	47.2 %
	Female	214	52.8 %
Age	Below 20	2	0.5 %
	20–29	133	32.8 %
	30–39	174	43.0 %
	40–49	72	17.8 %
	Above 50	24	5.9 %
Education	High school or below	1	0.2 %
	Three-year college	116	28.6 %
	Four-year college	206	50.9 %
	Graduate school or above	82	20.2 %
Frequency of use	Multiple times per day	89	22.0 %
	Once a day	84	20.7 %
	Multiple times per week	122	30.1 %
	Once a week	24	5.9 %
	Multiple times per month	36	8.9 %
	Once a month	28	6.9 %
	Less than once a month	22	5.4 %

Table A.3
Factor loadings and cross loadings in PLS.

	ANA	COA	CRA	LAS	LCS	LWS	PC	RB	SI
ANA1	**0.83**	0.15	0.25	0.29	0.32	0.21	0.08	0.14	−0.05
ANA2	**0.86**	0.18	0.24	0.31	0.26	0.23	0.08	0.12	−0.08
ANA3	**0.89**	0.26	0.28	0.32	0.37	0.28	0.05	0.19	−0.10
COA1	0.21	**0.85**	0.15	0.29	0.26	0.30	0.07	0.26	−0.10
COA2	0.19	**0.88**	0.21	0.30	0.27	0.23	0.10	0.18	−0.04
COA3	0.20	**0.86**	0.23	0.28	0.25	0.27	0.02	0.16	−0.08
CRA1	0.24	0.21	**0.79**	0.28	0.25	0.23	0.06	0.11	0.01
CRA2	0.23	0.18	**0.83**	0.26	0.24	0.17	0.01	0.08	−0.07

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Table A.3 (*continued*)

	ANA	COA	CRA	LAS	LCS	LWS	PC	RB	SI
CRA3	0.26	0.16	**0.84**	0.25	0.30	0.20	0.04	0.11	0.03
LAS1	0.30	0.26	0.27	**0.86**	0.21	0.28	−0.06	0.33	−0.14
LAS2	0.32	0.32	0.30	**0.88**	0.25	0.29	−0.05	0.28	–0.14
LAS3	0.30	0.29	0.28	**0.91**	0.23	0.31	0.00	0.37	−0.12
LAS4	0.33	0.29	0.27	**0.86**	0.21	0.33	−0.04	0.40	−0.15
LCS1	0.32	0.28	0.28	0.21	**0.87**	0.26	0.01	0.28	−0.10
LCS2	0.27	0.22	0.25	0.16	**0.84**	0.24	−0.03	0.22	−0.15
LCS3	0.33	0.23	0.26	0.25	**0.85**	0.30	−0.09	0.32	−0.21
LCS4	0.35	0.29	0.31	0.25	**0.87**	0.25	−0.05	0.23	−0.19
LWS1	0.27	0.26	0.20	0.29	0.26	**0.80**	−0.10	0.36	−0.13
LWS2	0.19	0.24	0.18	0.29	0.22	**0.82**	−0.11	0.34	−0.16
LWS3	0.21	0.26	0.22	0.24	0.25	**0.80**	−0.14	0.37	−0.26
LWS4	0.24	0.26	0.20	0.31	0.28	**0.85**	−0.12	0.31	−0.26
AU1	0.08	0.04	0.05	−0.04	−0.05	−0.15	**0.94**	−0.02	0.20
AU2	0.06	0.12	0.05	−0.04	−0.03	−0.08	**0.83**	−0.05	0.17
AU3	0.06	0.14	0.04	0.02	0.02	−0.03	**0.76**	0.04	0.14
RB1	0.15	0.20	0.13	0.35	0.28	0.41	−0.01	**0.88**	−0.08
RB2	0.13	0.22	0.10	0.36	0.29	0.36	−0.03	**0.91**	−0.10
RB3	0.19	0.20	0.10	0.34	0.25	0.35	−0.06	**0.89**	−0.07
SI1	−0.04	−0.08	−0.02	−0.16	−0.13	−0.22	0.15	−0.10	**0.81**
SI2	−0.10	−0.11	−0.03	−0.12	−0.20	−0.19	0.18	−0.11	**0.85**
SI3	−0.09	−0.03	0.03	−0.11	−0.16	−0.21	0.20	−0.03	**0.85**

Table A.4
Results of confirmatory factor analysis.

Constructs	indicators	Standardized	Cronbach's a	CR	AVE
Creative affordance	CRA1	0.79	0.75	0.76	0.67
	CRA 2	0.83			
	CRA 3	0.84			
Analysis affordance	ANA1	0.83	0.83	0.84	0.74
	ANA 2	0.86			
	ANA 3	0.89			
Communication affordance	COA1	0.85	0.83	0.83	0.75
	COA 2	0.88			
	COA 3	0.86			
Loss of worth-based self-esteem	LWS1	0.80	0.83	0.84	0.67
	LWS2	0.82			
	LWS3	0.80			
	LWS4	0.85			
Loss of competence-based self-esteem	LCS1	0.87	0.88	0.88	0.73
	LCS2	0.84			
	LCS3	0.85			
	LCS4	0.87			
Loss of authenticity-based self-esteem	LAS1	0.86	0.90	0.90	0.77
	LAS2	0.88			
	LAS3	0.91			
	LAS4	0.86			
Resistance behavior	RB1	0.88	0.87	0.87	0.80
	RB2	0.91			
	RB3	0.89			
Autonomy	AU1	0.94	0.85	0.78	0.72
	AU 2	0.83			
	AU 3	0.76			
Self-identity	SI1	0.81	0.78	0.78	0.70
	SI2	0.85			
	SI3	0.85			

Table A.5
Correlation coefficient matrix and square roots of the AVEs (shown as diagonal elements).

	ANA	COA	CRA	LAS	LCS	LWS	PC	RB	SI
ANA	0.86								
COA	0.23	0.87							
CRA	0.30	0.23	0.82						
LAS	0.36	0.33	0.32	0.88					
LCS	0.37	0.30	0.32	0.26	0.86				
LWS	0.28	0.31	0.25	0.35	0.31	0.82			
AU	0.08	0.07	0.05	−0.04	−0.05	−0.14	0.85		

(*continued on next page*)

Table A.5 (*continued*)

	ANA	COA	CRA	LAS	LCS	LWS	PC	RB	SI
RB	0.17	0.23	0.12	0.39	0.31	0.42	−0.04	0.89	
SI	−0.09	−0.09	−0.01	−0.16	−0.19	−0.25	0.21	−0.10	0.84

Note: CRA = Creative affordance; ANA = Analysis affordance; COA = Communication affordance; LAS = Loss of authenticity-based self-esteem; LCS = Loss of competence-based self-esteem; LWS = Loss of worth-based self-esteem; RB = Resistance Behavior; SI = Self-identity; AU = Autonomy.

Table A.6
Heterotrait–monotrait ratio of correlations (HTMT).

	ANA	COA	CRA	LAS	LCS	LWS	PC	RB	SI
ANA									
COA	0.27								
CRA	0.38	0.28							
LAS	0.41	0.38	0.39						
LCS	0.43	0.35	0.39	0.29					
LWS	0.33	0.37	0.31	0.40	0.36				
AU	0.09	0.13	0.07	0.05	0.05	0.13			
RB	0.20	0.27	0.15	0.44	0.35	0.49	0.05		
SI	0.11	0.11	0.07	0.19	0.23	0.30	0.24	0.12	

Note: CRA = Creative affordance; ANA = Analysis affordance; COA = Communication affordance; LAS = Loss of authenticity-based self-esteem; LCS = Loss of competence-based self-esteem; LWS = Loss of worth-based self-esteem; RB = Resistance behavior; SI = Self-identity; AU = Autonomy.

Table A.7
Weights for the formative construct.

Construct	Indicator	Weight	VIF
GAI identity threat	Loss of worth-based self-esteem	0.33***	1.16
	Loss of competence-based self-esteem	0.31***	1.12
	Loss of authenticity-based self-esteem	0.36***	1.15

*$p < 0.05$; **$p < 0.01$;.
*** $p < 0.001$.

Table A.8
Common method bias analysis.

Relationships	Research Model (without Marker Variable)	Research Model (with Marker Variable)
GAI identity threat → Resistance behavior	0.82***	0.82***
Creative affordance → GAI identity threat	0.25***	0.24***
Analysis affordance → GAI identity threat	0.24***	0.24***
Communication affordance → GAI identity threat	0.20***	0.20***
Marker → GAI identity threat		0.02
Marker → Resistance behavior		−0.05

*$p < 0.05$; **$p < 0.01$;.
*** $p < 0.001$.

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